

# **CM3070 Project Design**

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**ARIA – Activity Recognition & Inference Architecture: A Multi-Human Activity Detection & Role-Based Tracking System in a Shared Service Counter Environment**

**Template: CM3065 13.2 – Multi-human Detection and Activities Tracking in a Shared Space**

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# 1. Project Overview

ARIA (Activity Recognition & Inference Architecture) proposes a privacy-preserving, multimodal system that detects and tracks the real-time activities of multiple staff members operating simultaneously in a shared ticketing counter environment. The system identifies who is performing which activity at any given moment, using a combination of laptop-based computer vision, ambient audio analysis, and supplementary IoT sensing via an Arduino ESP8266 node. A secondary contribution is the production of a labelled, publishable dataset of counter-staff activities, the first of its kind in a shared, multi-occupant workplace domain.

The core motivation is the inadequacy of existing approaches in shared-service environments, such as transport ticketing counters, bank branches, and service desks, where operational decisions, such as when to deploy additional counter agents or when a manager's intervention is needed, currently rely on manual supervision or post-hoc CCTV review. Neither approach is efficient, scalable, or automated.

## 2. Domain and Users

### 2.1. Domain

ARIA operates in the domain of Human Activity Recognition (HAR), specifically at the intersection of multi-subject recognition, edge AI inference, and workplace operational monitoring. The physical deployment context is a shared service counter environment, a semi-public workplace where two distinct staff roles coexist in one space and perform simultaneous, potentially overlapping activities.

The domain problem is well-established in the literature. Arrotta et al. (2025) identify passive, camera-and-microphone-based multi-subject attribution in shared spaces as a frontier challenge, confirming that while single-subject HAR has achieved mature performance across sensing modalities, multi-subject HAR in shared occupancy remains open and complex. No existing commercial or academic system simultaneously provides role-aware attribution, passive ambient sensing, multimodal fusion, edge inference, and domain coverage of the service counter context.

### 2.2. Users

The system is designed for two primary user groups:

- Counter agents: staff stationed at the front, serving customers. Their tracked activities include serving (facing forward, leaning in), typing/processing (keyboard interaction), idle/waiting, and reaching/handling.
- Station managers: staff moving between the front and back office, supervising operations and handling administrative tasks. Their tracked activities include walking (zone transitions), desk work (seated, screen/paperwork), and in-conversation (facing another staff member).

## 3. Design Justification

### 3.1. Multimodal Sensing

The decision to fuse three sensor modalities, computer vision (skeletal keypoints), ambient audio (MAX9814), and IoT sensing (PIR motion + ultrasonic distance), is grounded in empirical evidence. Nakabayashi and Saito (2024) demonstrated that multimodal fusion achieves approximately 97% precision for activity classification on edge devices, outperforming single-sensor baselines, and that a compact model can perform this fusion locally without cloud offloading. Cristina et al. (2024) further established, through a scoping review of audio-video HAR systems in healthcare, that combining audio and video modalities significantly improves accuracy for activities with similar visual signatures, particularly the ambiguous pairs present in ARIA's activity taxonomy, such as a counter agent typing versus speaking on a phone.

Golipoor and Sigg (2024) validated the use of passive ambient sensors, rather than cameras alone, as a viable strategy for workplace activity recognition, and demonstrated that CNN-based classifiers effectively process heterogeneous non-visual sensor inputs. This supports ARIA's choice of a 1D-CNN as a candidate classifier for fused IoT features.

### **3.2. Edge-First, Privacy-by-Design Architecture**

All vision and ML processing runs locally on the inference host (laptop), with only statistical feature vectors, never raw video, transmitted to the cloud. This design responds directly to the tension between privacy and edge processing identified across the literature: both Li et al. (2023) and Cristina et al. (2024) acknowledge that cloud video processing raises significant data protection concerns, and the broader GDPR-compliant edge AI literature positions edge-first architectures as a regulatory necessity when processing individuals' visual and audio data in semi-public spaces.

Li et al. (2023) validated the hybrid edge-cloud paradigm – separating lightweight inference at the edge from resource-intensive model retraining in the cloud – demonstrating that multi-user recognition is feasible without centralised compute. ARIA maps directly onto this pattern: Tier 2 (laptop inference) mirrors Li et al.'s BSN inference layer, and Tier 3 (Google Colab retraining with Firebase sync) mirrors their cloud training layer.

### **3.3. Passive Ambient Sensing over Wearables**

ARIA deliberately avoids wearable sensors, despite their accuracy advantage in identity disambiguation, because wearables require active participant cooperation and are impractical in a naturalistic service counter environment. Golipoor and Sigg's RFID-based system and Li et al.'s BSN approach both achieve strong results but require tags or body sensors attached to individuals, an assumption that breaks down in real-world deployment. ByteTrack (Zhang et al., 2022), ARIA's chosen multi-person tracking backbone, bridges this gap by maintaining stable track IDs across occlusion events at 30FPS, providing identity continuity without wearables.

### **3.4. Dataset Contribution**

The secondary contribution, a labelled, publishable multimodal dataset, is justified on two grounds. First, Arrotta et al. (2025) identify the scarcity of multi-subject annotated datasets as a major bottleneck for progress in the field. Second, Moreaux et al. (2019), whose Kitchen20 methodology ARIA adopts (activity scripting, per-class labelling, data card documentation), demonstrated that domain-specific datasets become reference benchmarks that enable downstream research. ARIA expands the Kitchen20 template from a single-person, audio-only protocol to a multimodal, multi-occupant labelling problem.

## **4. System Architecture**

ARIA operates across three tiers:

### **4.1. Tier 1 – Edge Sensor Node (Arduino ESP8266)**

Three HC-SR501 PIR motion sensors and two HC-SR04 ultrasonic distance sensors are deployed across two named zones (Counter Zone A and Back Zone B), with a MAX9814 analogue microphone module on pin A0. The ESP8266 reads all sensors and transmits JSON payloads over serial USB to the edge inference host.

### **4.2. Tier 2 – Edge Inference Host (Laptop)**

A Python application subscribes to the ESP8266 serial stream and simultaneously captures webcam video and microphone audio. A YOLOv8n model detects and tracks persons using ByteTrack; a pose estimation model (MediaPipe or YOLOv8-pose) extracts skeletal keypoints. Combined with IoT sensor features, a 1D-CNN or Random Forest classifier identifies the activity class of each tracked individual in real time. Results are logged to a timestamped CSV dataset and visualised on a real-time Dash dashboard.

### 4.3. Tier 3 – Cloud Layer (Google Colab + Firebase)

Feature vectors, not raw video, are periodically synced to Firebase Realtime Database. Google Colab retrains the classifier on accumulated data and pushes updated model weights back to the inference host in an iterative loop.

### 4.4. Activity Classes by Role

Table 1: Activity Classes by Role

| Role            | Activity Classes                                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Counter Agent   | Serving (facing forward, leaning in); typing/processing (keyboard interaction); idle/waiting; reaching/handling |
| Station Manager | Walking (zone transition); desk work (seated, screen/paperwork); in-conversation (facing another staff member)  |

## 5. Key Technologies and Methods

Table 2: Key Technologies and Methods

| Technology/Method              | Role in ARIA                                                                                                                                                                                 |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| YOLOv8n                        | Real-time person detection; lightweight model suitable for edge deployment                                                                                                                   |
| ByteTrack (Zhang et al., 2022) | Multi-person identity tracking; associates low-confidence detection boxes to prevent fragmented tracklets under occlusion; 80.3 MOTA on MOT17 at 30FPS                                       |
| MediaPipe/YOLOv8-pose          | Skeletal keypoint extraction per tracked individual; joint positions and velocities used as classifier features                                                                              |
| 1D-CNN/Random Forest           | Activity classification from fused feature vectors (keypoints + audio amplitude + PIR/ultrasonic readings)                                                                                   |
| Arduino ESP8266<br>NodeMCU     | IoT hub reading HC-SR501 PIR sensors and HC-SR04 ultrasonic sensors, transmitting JSON over serial USB                                                                                       |
| MAX9814 Microphone<br>Module   | Ambient audio amplitude capture for discriminating visually ambiguous activity pairs                                                                                                         |
| Firebase Realtime Database     | Cloud storage of abstracted feature vectors; enables iterative model retraining without transmitting raw video                                                                               |
| Google Colab                   | Cloud-side classifier retraining; pushes updated model weights to the edge inference host                                                                                                    |
| Python/Plotly Dash             | Inference pipeline orchestration and real-time activity dashboard                                                                                                                            |
| Feature-level fusion           | Combining heterogeneous modality signals (skeleton keypoints, audio, IoT) into a unified vector before classification, as validated by Nakabayashi & Saito (2024) and Golipoor & Sigg (2024) |

## 6. Work Plan



Figure 1: ARIA Project Gantt Chart

The project spans approximately 19 weeks, structured across seven phases. Each phase is designed to be feasible within resource constraints (existing laptop hardware, low-cost Arduino components) and follows a build-then-validate progression.

Table 3: Work Plan

| Phase                               | Tasks                                                                                                         | Duration   | Contingency                                                                                                                                                                                      |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1: Setup & Hardware           | Deployment environment, GitHub repo, ESP8266 firmware, sensor wiring and serial testing                       | Weeks 1-2  | If ESP8266 serial communication is unstable, switch to USB direct UART; use simulated JSON payloads to unblock downstream development                                                            |
| Phase 2: Vision & Tracking Pipeline | YOLOv8n integration, ByteTrack tracking, MediaPipe/YOLOv8-pose keypoint extraction                            | Weeks 3-5  | If YOLOv8-pose is too slow on target hardware, fall back to MediaPipe Pose (lighter), or reduce inference resolution                                                                             |
| Phase 3: Audio & IoT Integration    | MAX9814 audio amplitude extraction, PIR/ultrasonic feature parsing, multimodal feature vector assembly        | Weeks 6-7  | If microphone module produces excessive noise floor, pre-process with bandpass filter; PIR can be replaced by zone-entry inference from vision alone                                             |
| Phase 4: Data Collection            | Activity scripting, participant informed consent, simulated environment recording sessions, manual annotation | Weeks 8-10 | If participant recruitment is slow, begin with researcher self-enactment (minimum viable: 2 roles, 2 participants); annotation tooling switched to CVAT if manual CSV annotation proves too slow |

|                                     |                                                                                                                                                        |             |                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 5: Classifier Training        | Feature vector construction, 1D-CNN and Random Forest training, hyperparameter tuning, Firebase/Colab cloud loop integration                           | Weeks 11-13 | If 1D-CNN overfits on small dataset, regularise with dropout/L2 or switch primary classifier to Random Forest; if Colab GPU quota is exhausted, train locally                                                                  |
| Phase 6: Evaluation & Testing       | Per-class F1 and confusion matrix analysis, leave-one-session-out cross-validation, ablation study, Dash dashboard integration, end-to-end system test | Weeks 14-16 | If F1 scores for a specific class are below threshold, review annotation quality and collect additional data for that class; dashboard can be simplified to static CSV export if real-time rendering is too resource-intensive |
| Phase 7: Write-up & Dataset Release | Final report writing, dataset data card documentation, Kaggle/Zenodo upload                                                                            | Weeks 17-19 | Dataset release delayed to post-submission if ethics documentation is incomplete; report sections written in parallel with Phase 6 to avoid bottleneck                                                                         |

## 7. Testing and Evaluation Plan

### 7.1. Quantitative Performance Metrics

The primary evaluation framework follows the protocols prescribed by Arrotta et al. (2025) for a credible multi-subject HAR system: per-class F1 score, confusion matrix analysis, and leave-one-session-out (LOSO) cross-validation. F1 score is preferred over accuracy because activity class distribution is inherently imbalanced and accuracy would inflate apparent performance by favouring the majority class.

The target thresholds, aligned with ARIA's proposed outcome tiers, are:

- Minimum (Pass): System detects and tracks at least two activity types across two simultaneous participants
- Good (Merit): Accurately detects 4+ activity types; full edge-cloud retraining loop demonstrated
- Outstanding: Per-class F1 metrics with ablation study results; publishable labelled dataset with full data card documentation

### 7.2. Ablation Study

An ablation study will evaluate the marginal contribution of each sensing modality by comparing classifier performance across configurations:

- vision only,
- vision + audio,
- vision + IoT, and
- full multimodal (vision + audio + IoT).

This directly responds to the three-way modality trade-off identified in the literature synthesis, validating whether each sensor stream contributes meaningfully to overall performance and quantifying the cost of removing any single modality.

### 7.3. Per-Role Attribution Accuracy

Since ARIA's distinguishing claim is role-aware attribution, and not just activity detection, evaluation must separately assess per-role accuracy, the ability to correctly assign the detected activity to the correct role (counter agent vs. station manager). Confusion between roles, even when the activity class is correctly identified, constitutes a system failure in the core use case.

### 7.4. Real-Time Functional Testing

End-to-end system testing will verify that the full pipeline, from sensor ingestion to inference to dashboard display, operates in real time (target: 33 ms or less per inference cycle, consistent with ByteTrack's 30 FPS throughput). This will be tested in the simulated counter environment under realistic occlusion conditions to replicate deployment stressors.

### 7.5. Ethical Compliance Verification

As the system processes visual and audio data of human participants, a compliance checklist will confirm that:

- all participants have provided written informed consent,
- ROI masking was applied before any frame was processed or stored,
- no raw video was retained beyond the local session, and
- the published dataset contains only anonymised sensor readings and activity labels,

in accordance with UK GDPR (Data Protection Act 2018).

### 7.6. Dataset Quality Evaluation

Following the Kitchen20 methodology (Moreaux et al., 2019), the dataset's quality will be assessed by:

- class balance across activity labels,
- inter-annotator agreement (Cohen's kappa, target of 0.8 or higher) if multiple annotators are used, and
- a benchmark comparison between the 1D-CNN and Random Forest classifiers to establish baseline performance figures for subsequent researchers.

### 7.7. Determining Success

The project will be considered to have achieved its primary aim if:

1. the system correctly attributes role-specific activities to the correct individual across simultaneous two-person sessions with a macro F1 of  $\geq 0.70$  for the merit tier or  $\geq 0.80$  for the outstanding tier,
2. the edge-cloud retraining loop completes without transmitting raw video to the cloud, and
3. the Dash dashboard displays activity labels in real time.

The secondary contribution (publishable dataset) will be considered achieved if the dataset is complete with a full data card and uploaded to Kaggle or Zenodo with participant consent.

## 8. References

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